Assignment Name:

Understanding the Problem:

1. How did you approach understanding the challenge?

   This challenge involves the reverse of previous challenges. The computer itself has to compute a random value and compare it to what the user inputs.

2. Were there any parts of the problem that you found confusing at first? If so, how did you resolve the confusion?

   None

Planning the Solution:

1. Did you create a plan or break the problem into smaller steps before coding?

I broke it into five parts: Generating the numbers, Generating the operation, Calculating the answer, Receiving input, and Comparing the answer to the user input.

2. How did you decide on which tools, algorithms, and data structures to use?

I decided on using the built in random package since it makes generating numbers easier and more readable.

Implementation:

1. Did you write the code in smaller pieces or attempt the entire solution at once?

   I attempted the entire solution at once.

2. How did you test your solution to ensure that it was working correctly?

I tested multiple combinations of values that I know the answer of.

Overcoming Challenges:

1. What part of the problem was the most difficult for you?
   The hardest part of this mastery problem was implementing division. Unlike other operations, division has multiple exceptions that have to be considered. For example, division is the only operation in which two integers can create a rational number meaning I have to change the type of the methods and variables handling the value as the superclass “Number” instead of “int”. Division by 0 is also mathematically unsolved meaning I have to create another exception for division where the dividing value cannot be 0.

2. How did you handle moments where you felt stuck or unsure what to do next?
   I asked ChatGPT

Learning:

1. Was there anything that you learned that you think will help you with future challenges?

   There are plenty of exceptions I have to consider when implementing division operations in the future.